Anomaly Segmentation for Road Scenes: A Comparative Study of Pixel- and Mask-Based Baselines

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Abstract

In this report we explore semantic segmentation and out-of-distribution (OoD) anomaly detection for road scenes. We compare two architectures: ERFNet [18], a pixel-based CNN, and EoMT [11], a mask-based Transformer built on a DINOv2 [17] backbone, with three checkpoints: Cityscapes [6], COCO [13], and a version fine-tuned from COCO to Cityscapes using LoRA [9] (rank r=8, α=16). To align the COCO model's 133-class output with the 19 Cityscapes classes, we define a manual semantic mapping function that assigns each COCO category to its nearest Cityscapes equivalent and sets IoU to zero for unmappable classes. For anomaly detection, we apply four post-hoc scoring methods (MSP [8], Max Logit [10], Max Entropy [3], and RbA [16]) on five road benchmarks, and test temperature scaling [7]. EoMT-Cityscapes achieves 81.7% mIoU and outperforms ERFNet across all anomaly benchmarks; LoRA fine-tuning brings the COCO model from 54.0% to 74.3% mIoU. LoRA fine-tuning allows improvement on FS Static [1] (+27.2% AuPRC) and FS LostFound [1] (+49.2%), but degrades FPR95 on RoadAnomaly21, revealing a trade-off between sensitivity and over-confidence. Temperature scaling has negligible effect, confirming the model is already well-calibrated at T=1.

The code is on Github : `https://github.com/ange237/MaskArchitectureAnomaly_CourseProject.git`

1. Introduction

Context of the project. Autonomous driving needs systems that can understand every pixel of a road while recognizing objects that the model did not see before during the training process.Therefore, there are two aspects involved in this process, semantic segmentation [18], where every pixel gets classified according to its class, and anomaly segmentation [2], detecting out-of-distribution (OOD) pixels (pixels from some classes not seen during training.) Both processes can be combined into panoptic segmentation [12], the process which combines semantic and instance segmentation. In the real world, unpredictable objects like animals, objects lying on the street, fallen cargo, etc., may occur at any time and not detect them is a potential risk of safety.

Literature foundations. There are two types of segmentation architectures: First, there are pixel-based CNNs like ERFNet, [18], which provides per-pixel score distribution by means of an efficient encoder-decoder architecture with residual factorisation layers. Second, mask-based Transformers, descendants of MaskFormer [4] and Mask2Former [5], which generate a set of binary masks with class labels, enabling unified handling of different kinds of segmentation: semantic, instance, and panoptic.EoMT [11] belongs to the second family of architectures by using DINOv2 [17] which is a Vision Transformer pre-trained on the large-scale image classification problem using self-supervised learning.
For anomaly detection, we prefer post hoc methods rather than retraining models. Maximum Softmax Probability (MSP) [8] assumes that the low predicted confidence corresponds to an OOD. Energy scoring technique [15] uses log-sum-exp of logits as a free energy and shows better OoD calibration than MSP. Max Logit [10]considers the maximum raw logit value. For mask-based architectures RbA [16] takes the sum of class probabilities in mask proposals and reports OOD pixels with low total probability for the known classes. Temperature scaling [7] is a post hoc calibration approach, in which logits get scaled by a scalar T before the application of the softmax function.

2. Methodology

2.1. Mapping Function

The output of the EoMT-COCO model (EoMT model trained on COCO dataset) provides predictions in 133

classes/categories while the cityscape test dataset is based on the 19 categories. In order to obtain the mIoU score, we use a deterministic mapping function $f : \mathcal{C}_{\text{COCO}} \to \mathcal{C}_{\text{City}} \cup \{\varnothing\}$ which takes an element from a COCO category and returns its closest match in the Cityscapes domain, if such a mapping exists, else the empty set $\varnothing$ when no correspondence exists. If the predicted label is $\hat{y}_i \in \mathcal{C}_{\text{COCO}}$ at index i,then the transformed prediction is:

$$\tilde{y}_i = \begin{cases} f(\hat{y}_i) & \text{if } f(\hat{y}_i) \neq \varnothing \\ \textit{ignore} & \text{otherwise} \end{cases} \tag{1}$$

Pixels marked as non-mappable are ignored during the IoU calculation for all classes, which means that unmappable COCO classes will not unfairly increase or decrease the score of any class in favor of either model. On the other hand, classes from Cityscapes that have no equivalent COCO classes, such as pole and rider, will never be found in the COCO set and will therefore give an IoU = 0 score, which is a fair penalty.
The remapping is done by manually comparing names and meaning of the categories. Example remappings are : *car→car*, *person→person*, *stop sign→traffic sign*. The entire remapping pipeline, including use of mIoU script, will be identical for EoMT-Cityscapes model and EoMT-COCO model.

2.2. Post-Hoc Anomaly Scoring Methods

Given a segmentation model, The anomaly score s_i of each pixel is generated using the model's predictions without needing to train the model again. The larger the value of the score, the higher the likelihood that the object belongs to an anomaly class.
MSP [8]: $s_i = 1 - \max_c p_{ic}$, where p_{ic} is the softmax probability for class c. Low maximum confidence is treated as an OoD signal.
Max Logit [10]: $s_i = -\max_c z_{ic}$, where z_{ic} is the raw logit. This avoids the saturation of the softmax function.
Max Entropy [3]: $s_i = \mathcal{H}(p_i) = -\sum_c p_{ic} \log p_{ic}$. High entropy means the model is uncertain across all classes.
RbA [16]: applicable only to mask-based models (EoMT). For pixel i, let $\{(\mathbf{m}_k, \mathbf{q}_k)\}$ be the set of predicted masks and their class logits. The anomaly score is $s_i = -\sum_k m_{ki} \cdot \max_c q_{kc}$, which flags pixels where the total mask-weighted class confidence is low.
Temperature scaling [7]: logits are rescaled as $\tilde{z}_{ic} = z_{ic}/T$ before applying any scoring method. We sweep $T \in \{0.5, 0.75, 1.0, 1.1\}$ with MSP on EoMT-Cityscapes.

2.3. Fine-Tuning with LoRA

The EoMT-COCO model has good general capabilities, but it has not been trained on road scene images before. We fine-tune this model on Cityscapes using **LoRA** [9], which injects trainable low-rank matrices $\Delta W = BA$ into the attention projection layers of the ViT backbone, with rank r=8 and scaling α=16. In the training process, only ΔW is updated during training; the original weights W_0 remain frozen. The classifier needs to be modified in order to match (133→19 classes) and trained from scratch . Training uses AdamW with automatic mixed precision (AMP) on the Cityscapes training set. This approach keeps the number of trainable parameters small while allowing the model to specialise for road-scene semantics.

2.4. Evaluation Metrics

The **mIoU** measure is used to determine the success of semantic segmentation. For each class c, the IoU is calculated using the formula:$\text{IoU}_c = \text{TP}_c/(\text{TP}_c + \text{FP}_c + \text{FN}_c)$. The mean of IoU is calculated for all 19 classes of the Cityscape dataset.
The **AuPRC** (Area under the Precision-Recall Curve,) measures the anomaly detection quality across all thresholds. It is useful when the anomalies are very few. Larger value is better.
The **FPR95** (False Positive Rate at 95% True Positive Rate) measures the fraction of in-distribution pixels that are wrongly considered as anomalous by the detector while maintaining a true positive rate of 95%. A lower FPR95 means fewer false alarms.

3. Experimental Results

3.1. Semantic Segmentation — mIoU

The result of the per-class IoU of all EoMT models (Cityscape ,COCO and Finetuned) on the Cityscape validation dataset are reported in the Table 1. The mIoU of EoMT-Cityscapes is **81.7%**. On the other hand, EoMT-COCO has an mIoU of just **54.0%** since three classes of the Cityscapes dataset are not available in the COCO dataset(*pole* (0.0%), *rider* (0.0%), and *traffic sign* (4.4%) score near zero). The finetuned EoMT-COCO model gains 20.3 percentage points and reaches **74.3%** mIoU,
Figure 1 shows a qualitative comparison on Cityscape validation dataset images. The Cityscapes model produces a clean semantic map with correct road labels. The COCO model map uses different colors and fails to predict road-specific categories.

3.2. Anomaly Detection Baselines

We evaluate our models on five benchmarks: SMIYC Road Anomaly21 (RA-21) and Road Obstacle21 (RO-21) [2], Road Anomaly (RA) [14], FS LostFound (FSLF), and FS Static (FSS). Table 2 reports AuPRC / FPR95 for all models and methods. Figure 2 summarises AuPRC results for MSP across EoMT checkpoints.
Important observations:

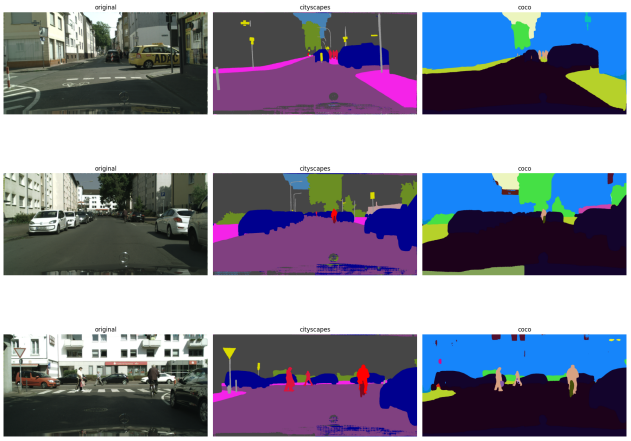


Figure 1. Predictions on a Cityscapes val image. *Left*: input. *Centre*: EoMT-Cityscapes model . *Right*: EoMT-COCO model .

Table 1. Per-class IoU (%) on Cityscapes validation. Red = class absent from COCO. City = EoMT-Cityscapes, COCO = EoMT-COCO, FT = LoRA fine-tuned.

Class	City	COCO	FT
road	98.4	94.0	97.8
sidewalk	87.4	64.7	83.1
building	94.1	78.8	92.3
wall	66.1	45.8	60.0
fence	65.5	48.1	57.7
pole	71.0	0.0	60.2
traffic light	75.0	60.4	61.4
traffic sign	82.1	4.4	74.4
vegetation	93.0	87.2	91.6
terrain	66.6	52.4	65.9
sky	95.5	89.7	94.6
person	85.4	67.2	79.9
rider	71.1	0.0	56.0
car	95.5	84.2	94.0
truck	81.8	29.3	77.5
bus	90.3	71.1	74.5
train	77.4	13.6	57.1
motorcycle	74.4	63.6	58.3
bicycle	81.3	72.2	75.0
mIoU	**81.7**	**54.0**	**74.3**

- EoMT-Cityscapes outperforms ERFNet on all datasets (e.g., 94.3% vs. 3.0% AuPRC on RO-21 with Max Entropy).
- EoMT-COCO fails on road benchmarks (FPR95 up to 100%) due to domain mismatch: the model has no concept of “normal road scene”.
- LoRA fine-tuning model gives large gains on FSS (+27.2% with MSP) and FSLF (+49.2% with Max Entropy), but FPR95 on RA-21 rises to 98.9%.

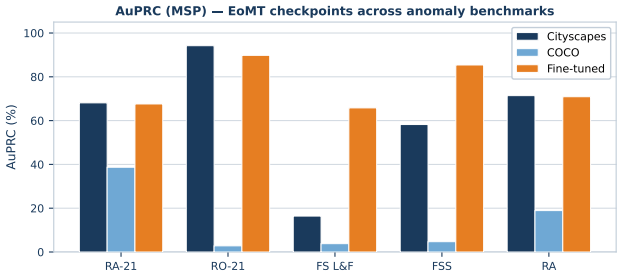


Figure 2. AuPRC (%) with MSP for each EoMT checkpoint across five benchmarks.

- RbA is competitive on RO-21 for Cityscapes (94.0% AuPRC, 0.3% FPR95) but collapses on RA-21 (FPR95 97.3%).

3.3. Temperature Scaling

Table 3 shows the impact of temperature on EoMT-Cityscapes with MSP. Differences across all T values are smaller than 0.2% AuPRC and 0.3% FPR95. The best temperature is T=1.1 on most datasets, but the gain is very low.

4. Discussion

Why EoMT-Cityscapes dominates. EoMT-Cityscapes clearly dominates in terms of performance. It reaches 81.7% mIoU and outperforms ERFNet on anomaly dataset(for example 94.3% vs. 3.0% AuPRC on RO-21). This difference can not only be explained by the model size of the architecture. The key factor is the features representation. DINOv2 [17] features are produced via self-supervised learning on a large and diverse collection of images, producing representations that generalise well and can easily capture any anomaly. When an anomalous object appears, its DINOv2 features are likely to differ from those of any known class, making it easier for any post-hoc scoring method to detect it. ERFNet features, by contrast, are only trained for classification on Cityscapes. Moreover, the predictions made by EoMT based on masks give much better results than the per-pixel softmax of ERFNet.

Why EoMT-COCO fails on road benchmarks. The post-hoc method assumes that the model has a good understanding of the definition of ”normal”. Since the EoMT-COCO model has been trained on natural images from the COCO dataset and has never seen a road scene,he doesn’t really know what it is. That’s why its uncertainty is much higher everywhere on road images (for both anomalous and in-distribution pixels). This makes it nearly impossible to distinguish anomalies from background, which explains the FPR95 values near 100% on RO-21 and FSS. This result shows the limitation of post-hoc methods: they strongly depend on the quality of

Table 2. Anomaly detection baselines. AuPRC (%,↑) / FPR95 (%,↓). **Bold** = best AuPRC per dataset and model group. "—" = not applicable.

Model	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AP	FPR	AP	FPR	AP	FPR	AP	FPR	AP	FPR
ERFNet	MSP	—	29.1	62.6	2.7	65.2	1.7	50.6	7.5	41.8	12.4	82.6
	Max Logit		**38.3**	59.4	**4.6**	48.4	**3.3**	45.5	**9.5**	40.3	**15.6**	73.3
	Max Entropy		31.0	62.7	3.0	65.9	2.6	50.2	8.8	41.5	12.7	82.8
EoMT Cityscapes	MSP	81.7	68.1	30.4	94.2	0.4	16.4	13.0	58.2	43.5	71.4	15.5
	Max Logit		67.5	31.6	94.2	0.4	16.4	12.8	58.4	46.9	70.8	15.1
	Max Entropy		**68.3**	30.6	**94.3**	0.3	**18.9**	12.8	56.9	43.8	**74.0**	14.7
	RbA		64.5	97.3	94.0	0.3	16.5	10.2	**60.2**	72.2	70.2	14.2
EoMT COCO	MSP	54.0	38.7	77.5	2.8	100	3.8	97.2	4.7	99.5	18.9	91.9
	Max Logit		39.0	77.0	2.7	100	3.8	97.2	**4.7**	99.5	19.0	91.6
	Max Entropy		**43.1**	63.6	8.3	100	**3.8**	96.9	4.6	99.3	**21.5**	87.9
	RbA		22.0	77.2	**51.8**	100	0.7	97.5	2.5	96.6	17.8	84.7
EoMT Fine-tuned	MSP	74.3	67.6	98.9	89.8	0.7	65.8	38.2	**85.4**	23.3	70.9	71.4
	Max Logit		**67.6**	99.1	89.9	0.7	66.5	51.6	85.1	42.7	70.0	72.3
	Max Entropy		67.6	98.9	**90.6**	0.5	**68.1**	48.7	85.3	56.2	**71.1**	69.8
	RbA		54.4	96.8	88.8	32.3	67.2	79.9	78.1	83.7	62.5	80.0

Table 3. Temperature scaling on EoMT-Cityscapes (MSP). AuPRC (%,↑) / FPR95 (%,↓). Blue = best T per dataset.

T	mIoU	RA-21		RO-21		FS Static	
		AP	FPR	AP	FPR	AP	FPR
1.0	80.6	68.1	30.4	94.2	0.4	58.2	43.5
0.5	80.6	67.9	30.4	94.2	0.4	58.2	43.5
0.75	80.6	68.0	30.4	94.2	0.4	58.2	43.5
1.1	80.6	**68.1**	30.4	**94.2**	0.4	**58.2**	43.5

the underlying model's in-distribution representation.

Why fine-tuning helps but introduces over-confidence on RA-21. After LoRA fine-tuning on Cityscapes, the model significantly improve going from 54.0% to 74.3% mIoU and learns a strong prior for road scenes. This dramatically reduces false positives on FSS and FSLF (datasets with static objects on urban roads), because the model now recognizes road-class labels for in-distribution pixels with great certainty. However, RoadAnomaly21 [2] contains more diverse anomalies that appear in varied environments and lighting conditions, some of which may resemble in-distribution objects after fine-tuning. The model becomes over-confident on these scenes, predicting a known class where it should express uncertainty, which raises FPR95 to 98.9%. This is an overfitting to distribution effect: the model becomes better at recognising "normal" but worse at recognising what it does not know.

Why RbA is unstable. RbA [16] ranks pixels by aggregating mask-level class confidence. On RO-21, where anomalies are small and clearly separated from the road surface, this works extremely well (0.3% FPR95 for EoMT-Cityscapes). On RA-21, however, anomalies are larger and more varied; the mask proposals may partially cover them with high confidence for a road class, which causes RbA to assign a low anomaly score and produces a collapsed FPR95 of 97.3%. This shows that RbA's behaviour is strongly dataset-dependent.

Why temperature scaling does not help. Temperature scaling [7] only rescales the confidence distribution; it can't change which pixel has the highest score relative to others. If the model is already well calibrated (like EoMT-Cityscapes appears to be, given its strong MSP results at $T{=}1$), there is no miscalibration to correct. The negligible differences across $T \in \{0.5, 0.75, 1.0, 1.1\}$ confirm that calibration is not the bottleneck here.

Methodological limitations. Our mapping function is manually defined and uses name matching, which introduces subjective choices. For example, the COCO category *stop sign* is mapped to Cityscapes *traffic sign*, which is an approximative mapping. Furthermore, the fine-tuned model's mIoU is not directly comparable to the provided EoMT-Cityscapes checkpoint because training hyperparameters (number of epochs, learning rate schedule) were not fully optimised due to compute constraints. Finally, the qualitative figure uses a single image, which may not be representative of the full

validation set.

5. Conclusion

The most successful model is EoMT-Cityscapes with Max Entropy, which achieves 94.3% AuPRC on RO-21 and 68.3% on RA-21. However the main limitation is that all post-hoc methods(MSP, Max Logit, Max Entropy, and RbA [16] are sensitive to domain mismatch and over-confidence after fine-tuning, meaning that future work should focus on joint training objectives that explicitly regularise uncertainty on in-distribution regions while remaining sensitive to OoD pixels.

6. Acknowledgements

We declare that no AI generative tools were used to write this report.
We declare the use of Generative AI tools for the development of the submitted project code, as detailed below:
Name and version: claude code
Publisher: Anthropic
Detailed description of use: The tool was used to assist in debugging specific functions and optimizing code syntax. All suggestions provided by the AI were subject to human supervision; we systematically reviewed, tested, and validated the output against our project requirements. Final decision-making regarding the code architecture and the interpretation of experimental results remained exclusively with the project team.

7. Revision Notes

In order to fix the problem mentioned in the feedback we replaced or deleted the wrong references (for instance we deleted the reference 11 of the old report) and we linked each reference with the correct term in the report. The order and number of references were also modified due to the modifications. We added some new references such as Cityscapes, COCO, Max Entropy, Max Logit and Semantic segmentation. We substituted some references with more precise ones (We subtituted the references of MSP and EoMT). The remaining references have not been modified but the numbering of references changed due to the modification of the report.

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